RAGHAVENDER REDDY BADUGULA

Technical Project Manager, Product Architect & Operations Lead | MSc Advanced Mechanical Engineering (Cranfield University)

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PROFESSIONAL SUMMARY

Results-driven Technical Project Manager, Product Architect, and Operations Lead with an MSc in Advanced Mechanical Engineering from Cranfield University and 3+ years of experience directing high-stakes project delivery, localization workflows, and cross-functional teams (National Testing Agency, TCS iON, IBPS, Smashburger UK). Skilled in Project Charters, Quantitative Risk Matrices (PI Scoring), Double Diamond design thinking, Gantt scheduling, and UK/ISO regulatory compliance. Bridges domain operations with AI-augmented product engineering—conceiving product logic and orchestrating AI coding workflows to ship end-to-end software tools including Polyglot Engine and IndiCAT.

CORE TECHNICAL & MANAGEMENT SKILLS

- **Project & Programme Management:** Project Charter & Scope Governance, Quantitative PM Risk Assessment Matrices (Probability-Impact Scoring & Mitigation), Multi-Phase Gantt Scheduling, Double Diamond Design Thinking, Quality Assessment Score Matrices, Stakeholder Mapping, Agile & Phased Delivery, SLA & Confidentiality Compliance.
- **Product Architecture & AI-Assisted Delivery:** Product Vision & Systems Logic Design, End-to-End AI-Augmented Software Prototyping & Execution, Localization & CAT Workflow Automation (IndiCAT, Polyglot Engine), GitHub Actions CI/CD, Playwright Visual QA, Python, Advanced Excel & Data Analytics.
- **Engineering & Standards Compliance:** UK/EU & ISO Standards (BS EN 12464-1 Lighting, ISO 3382-3 Acoustics, BS EN 1729-1 Ergonomics, BS 8300 Accessibility, COSHH Lab Safety, DNV Offshore Standards), Circular Economy & Sustainable Design, SolidWorks CAD, ANSYS Fluent/ICEM CFD, Abaqus FEA, MATLAB.
- **Operations & Team Leadership:** Cross-Functional Vendor & Freelancer Coordination, UK Frontline Shift & Inventory Management, Multi-Language Exam Operations (JEE Mains, CUET, IBPS), Technical Documentation & Executive Reporting.

EDUCATION & QUALIFICATIONS

MSc Advanced Mechanical Engineering | Cranfield University, UK — Oct 2024 – Sep 2025

- Key Modules: Engineering Design & Project Management (63%), Component Design (73%), Fluid Mechanics & Loading (74%), Structural Integrity (69%), Industrial Group Project with Nissan (66%), MSc Thesis (62%), Assessing Risk & Failure.

B.Tech Mechanical Engineering (Mechatronics) | MGIT, Hyderabad, India — Aug 2016 – Nov 2020

- GATE 2022 Qualified: Smart Manufacturing & Advanced Production Systems.

PROJECT MANAGEMENT & OPERATIONAL LEADERSHIP EXPERIENCE

Assistant Project Manager / Operational Coordinator | National Testing Agency (NTA), TCS iON & IBPS Vendors — Jun 2021 – Sep 2024

- Directed end-to-end localization, translation, validation, and data entry operations for India's largest high-stakes national entrance and banking examinations (JEE Mains, CUET 2023, IBPS, TCS iON) under immovable national deadlines.
- Managed cross-functional remote and on-site teams of freelance translators, subject matter experts (SMEs), and typists across multiple regional languages; allocated daily workloads, tracked milestone burn-down, and resolved bottlenecks.
- Enforced zero-defect quality assurance, mock-test validation, objection handling, and 100% compliance with strict high-security confidentiality protocols.

Team Lead / Operational Specialist | Smashburger, Milton Keynes, UK — Nov 2024 – Present

- Selected for Team Lead progression based on frontline operational excellence; managing daily shift execution, staff scheduling, inventory control, food safety auditing, and customer service KPIs in a fast-paced UK environment.

Freelance Localization Specialist & Interpreter | Netflix, Amazon Prime & UK Corporate Clients — 2018 – 2021

- Delivered high-accuracy English–Telugu subtitle localization for major streaming releases (including NCIS) and provided remote cross-cultural interpretation for UK-based legal and business clients.

AI-AUGMENTED PRODUCT ARCHITECTURE & SOFTWARE PROJECTS

Polyglot Engine — Autonomous Multilingual UI Scanner & Verified Repair CLI | github.com/Raghavender1234/polyglot-engine — Product Architect (2026)

- Identified a critical gap between string-level localization tools and pixel-level UI breakage; conceived and architected a 6-layer developer CLI that scans live DOMs (Playwright), diagnoses root causes (HarfBuzz text shaping, BiDi glitches, rigid CSS), and generates verified source patches.
- Authored the complete system logic and directed end-to-end implementation, automated test suites (pytest), interactive HTML diff reporting, and GitHub Action CI/CD integration using AI-assisted software development workflows.

IndiCAT — Desktop CAT & AI Subtitle Localization Platform | indicat.co.in — Founder & Architect (2024 – Present)

- Designed and shipped a Windows desktop Computer-Assisted Translation (CAT) and subtitle editor application integrating local and cloud LLMs, Progressive Web App (PWA) sync, and WhatsApp/Telegram bot workflows built end-to-end with AI coding assistance.

UK Food Store Rota Optimizer & Multi-Tenant AI Learning Management System (LMS) — Systems Logic Designer

- Designed constraint-based scheduling logic for UK retail shift optimization (uk-food-store-rota-optimizer) and architected a multi-tenant AI LMS with automated content scraping and exam generation pipelines.

CRANFIELD UNIVERSITY ENGINEERING & PROJECT MANAGEMENT PORTFOLIO

Project Manager — Stafford Cripps SC3 Smart Classroom Modernization Proposal | Cranfield University Mar 2025 – May 2025

- Led project planning using the Double Diamond Design Thinking framework; authored the Project Charter, stakeholder register, circular-economy furniture reuse case, and UK/ISO regulatory specification (BS EN 12464-1 LED, ISO 3382-3 acoustics, BS EN 1729-1 ergonomics, BS 8300 accessibility).
- Constructed a 15-item quantitative PM Risk Assessment Framework (driving Probability-Impact scores down from 25 to ≤9 via pre-ordering, UPS backup, and off-hour phased installation), Quality Assessment Score Matrix, and 11-week Gantt schedule.

Product Design & Prototyping — Smart & Sustainable Shower Recirculation System | Cranfield University (73%) 2025

- Conducted stakeholder influence mapping (40% end-users, 20% installers, 20% manufacturers), defined functional/non-functional requirements, modeled the assembly in SolidWorks, and 3D-printed a physical prototype cutting pre-bath cold water waste by 50–70%.

R&D Coordination — Lead-Free Semiconductor Solar Paint (Industrial Group Project with Nissan) | Cranfield 2024 – 2025

- Coordinated experimental scheduling, PhD mentor liaison, and technical report synthesis for a novel photovoltaic paint dispersing bismuth methyl iodide quantum dots (1.8–2.3 eV) in a TiO_2 matrix on galvanized automotive steel across 48-hour conductivity cycles.

Lead Researcher — High-Temperature Molten Salt Corrosion of WAAM SS316L for CSP TES (MSc Thesis) 2024 – 2025

- Designed and executed a 500-hour experimental test matrix (open vs. covered-lid crucibles at 600°C in $10NaNO_3$–$78KNO_3$–$12KCl$), authoring full COSHH risk assessments and integrating 30-position 3D optical profilometry, 10^{-5} g gravimetry, and SEM-EDS elemental mapping.

Computational Simulation & Risk Analysis Suite (Abaqus FEA, ANSYS ICEM CFD & Fluent, MATLAB BEM) 2024 – 2025

- Delivered technical simulation reports on Abaqus FEA offshore wind monopile buckling (23,200 elements, 5 modes), ANSYS ICEM CFD & Fluent urban VAWT NACA 0015 biomimetic optimization (104k-node mesh independence), and MATLAB/XFOIL HAWT BEM rotor & Airy wave kinematics (74%).